INCH - POUND

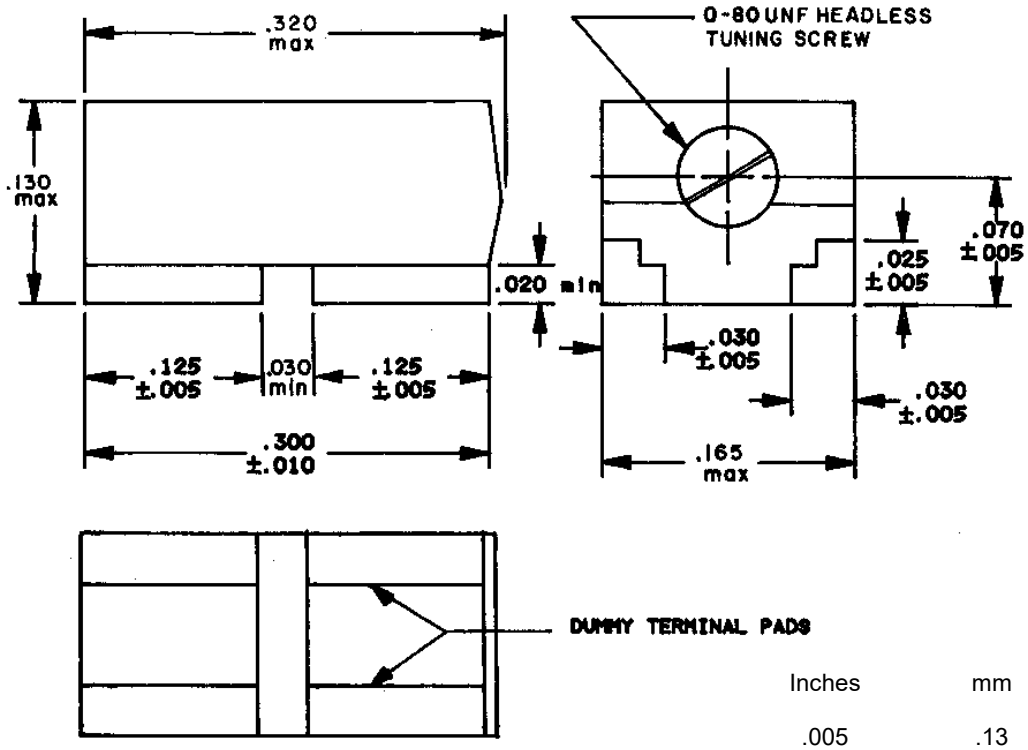
MIL-PRF-83446/19B
w/AMENDMENT 1
2 September 2019
SUPERSEDING
MIL-PRF-83446/19B
25 August 2006

PERFORMANCE SPECIFICATION SHEET

COILS, RADIO FREQUENCY, VARIABLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the products described herein shall consist of this specification and MIL-PRF-83446.




| Inches | mm |
|---|---|
| .005 | .13 |
| .010 | .25 |
| .020 | .51 |
| .025 | .64 |
| .030 | .76 |
| .070 | 1.78 |
| .130 | 3.30 |
| .165 | 4.19 |
| .300 | 7.62 |
| .320 | 8.13 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 1. Variable, chip coil.

AMSC N/A

FSC 5950

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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Case: Epoxy.

Weight: 0.5 gram, maximum.

Turning torque: .05 ounce-inches minimum, .50 ounce-inches maximum.

Operating temperature range: -55°C to +125°C.

Temperature rise (at 90°C): 35°C.

Maximum operating temperature: 125°C.

Altitude: 70,000 feet.

Dielectric withstanding voltage: MIL-STD-202-301, test voltage 700 volts rms.

Barometric pressure: MIL-STD-202-105, test condition C, (70,000 feet), test voltage 200 volts rms.

Electrical characteristics (initial): See table I.

Electrical characteristics (final): See table II.

Part or Identifying Number (PIN): The part number shall be in the following form.

| M83446/19- | 04 | B |
|---|---|---|
| | Sequentially assigned dash Numbers (see table I) | Termination finish (see MIL-PRF-83446) |



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TABLE I. Electrical characteristics (initial).

| Dash number | Inductance (μH) | | | Q min over L range | Q (typ) (at L nom) | Test Freq-uency (MHz) 1/ | Self resonant frequency (min) MHz | DC Resistance At 25 °C (max) ohms | Current (max) mA 2/ | Incre-mental current 3/ |
|---|---|---|---|---|---|---|---|---|---|---|
| | Min | Nom | Max | | | | | | | |
| 01 | 0.010 | 0.013 | 0.016 | 50 | 60 | 200 | 1200 | 0.03 | 250 | ---- |
| 02 | 0.018 | 0.022 | 0.025 | 50 | 60 | 200 | 1050 | 0.07 | 250 | ---- |
| 03 | 0.036 | 0.044 | 0.051 | 40 | 50 | 100 | 740 | 0.13 | 250 | ---- |
| 04 | 0.049 | 0.065 | 0.076 | 40 | 50 | 100 | 600 | 0.20 | 250 | ---- |
| 05 | 0.071 | 0.085 | 0.100 | 30 | 40 | 25 | 530 | 0.32 | 250 | ---- |
| 06 | 0.080 | 0.105 | 0.130 | 30 | 40 | 25 | 525 | 0.34 | 200 | ---- |
| 07 | 0.107 | 0.129 | 0.150 | 30 | 40 | 25 | 500 | 0.35 | 200 | ---- |
| 08 | 0.120 | 0.150 | 0.180 | 30 | 40 | 25 | 470 | 0.41 | 180 | ---- |
| 09 | 0.157 | 0.189 | 0.220 | 30 | 40 | 25 | 400 | 0.45 | 180 | ---- |
| 10 | 0.193 | 0.232 | 0.270 | 30 | 40 | 25 | 350 | 0.50 | 180 | ---- |
| 11 | 0.236 | 0.283 | 0.330 | 30 | 40 | 25 | 320 | 0.60 | 140 | ---- |
| 12 | 0.230 | 0.300 | 0.370 | 30 | 40 | 25 | 275 | 0.62 | 140 | ---- |
| 13 | 0.336 | 0.403 | 0.470 | 30 | 40 | 25 | 255 | 0.65 | 100 | ---- |
| 14 | 0.400 | 0.480 | 0.560 | 30 | 40 | 25 | 230 | 0.75 | 100 | ---- |
| 15 | 0.485 | 0.583 | 0.680 | 30 | 40 | 25 | 190 | 0.85 | 100 | ---- |
| 16 | 0.565 | 0.660 | 0.760 | 30 | 40 | 25 | 170 | 0.88 | 100 | ---- |
| 17 | 0.586 | 0.703 | 0.820 | 30 | 40 | 25 | 150 | 0.90 | 100 | ---- |
| 18 | 0.616 | 0.783 | 0.950 | 30 | 40 | 25 | 145 | 0.95 | 100 | ---- |
| 19 | 0.666 | 0.833 | 1.00 | 25 | 35 | 25 | 140 | 0.98 | 100 | ---- |
| 20 | 1.00 | 1.25 | 1.50 | 25 | 35 | 7.9 | 110 | 1.0 | 100 | ---- |
| 21 | 1.47 | 1.84 | 2.20 | 25 | 35 | 7.9 | 100 | 1.4 | 80 | ---- |
| 22 | 1.80 | 2.25 | 2.70 | 25 | 35 | 7.9 | 89 | 1.6 | 75 | ---- |
| 23 | 2.20 | 2.75 | 3.30 | 25 | 35 | 7.9 | 77 | 1.8 | 65 | ---- |
| 24 | 3.13 | 3.92 | 4.70 | 25 | 35 | 7.9 | 49 | 2.2 | 60 | ---- |
| 25 | 3.87 | 4.84 | 5.80 | 25 | 35 | 7.9 | 38 | 2.8 | 58 | ---- |
| 26 | 4.53 | 5.67 | 6.80 | 25 | 35 | 7.9 | 36 | 3.0 | 55 | ---- |
| 27 | 5.47 | 6.84 | 8.20 | 25 | 35 | 7.9 | 21 | 3.0 | 55 | ---- |
| 28 | 6.00 | 7.50 | 10.00 | 25 | 35 | 2.5 | 20 | 2.8 | 55 | 50 |
| 29 | 6.50 | 9.75 | 13.00 | 25 | 35 | 2.5 | 19 | 3.0 | 53 | 45 |

See footnotes at end of table.

TABLE I. Electrical characteristics (initial) - Continued.



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| Dash number | Inductance (μH) | | | Q min over L range | Q (typ) (at L nom) | Test Freq-uency (MHz) 1/ | Self resonant frequency (min) MHz | DC Resistance At 25 °C (max) ohms | Current (max) mA 2/ | Incre-mental current 3/ |
|---|---|---|---|---|---|---|---|---|---|---|
| | Min | Nom | Max | | | | | | | |
| 30 | 8.00 | 12.00 | 16.00 | 25 | 35 | 2.5 | 16 | 3.4 | 50 | 40 |
| 31 | 10.00 | 15.00 | 20.00 | 25 | 35 | 2.5 | 14 | 3.5 | 43 | 37 |
| 32 | 11.00 | 16.50 | 22.00 | 25 | 35 | 2.5 | 12 | 4.0 | 43 | 36 |
| 33 | 15.00 | 22.50 | 30.00 | 25 | 35 | 2.5 | 11 | 4.5 | 40 | 32 |
| 34 | 18.00 | 27.00 | 36.00 | 25 | 35 | 2.5 | 10 | 5.0 | 40 | 31 |
| 35 | 23.00 | 35.00 | 47.00 | 25 | 35 | 2.5 | 6.5 | 5.7 | 39 | 29 |
| 36 | 28.00 | 42.00 | 56.00 | 25 | 35 | 2.5 | 6.3 | 7.0 | 35 | 28 |
| 37 | 34.00 | 51.00 | 68.00 | 25 | 35 | 2.5 | 6.2 | 8.0 | 30 | 26 |
| 38 | 37.00 | 56.00 | 75.00 | 25 | 35 | 2.5 | 5.9 | 9.0 | 30 | 25 |
| 39 | 43.00 | 65.00 | 86.00 | 25 | 35 | 2.5 | 5.6 | 10.0 | 28 | 23 |
| 40 | 50.00 | 75.00 | 100 | 25 | 35 | 2.5 | 5.3 | 12.0 | 27 | 20 |
| 41 | 60.00 | 90.00 | 120 | 20 | 30 | 0.79 | 4.4 | 14.0 | 25 | 20 |
| 42 | 75.00 | 112.5 | 150 | 20 | 30 | 0.79 | 4.0 | 18.0 | 22 | 20 |
| 43 | 235 | 353 | 470 | 20 | 30 | 0.79 | 2.1 | 30.0 | 16 | 16 |
| 44 | 340 | 510 | 680 | 20 | 30 | 0.79 | 2.1 | 35.0 | 15 | 15 |
| 45 | 500 | 750 | 1000 | 20 | 30 | 0.79 | 1.1 | 75.0 | 11 | 11 |

1/ Test frequency range 0.25 through 25 MHz: For electrical characteristics measurements, use the HP260-A Q Meter, HP4342A, HP250B RX Meter, or equivalent with the TF-405 test fixture, or equivalent (see figure 2). Fixture inductance (approximately 0.038 μH) and residual Q-meter inductance (approximately 0.01 μH) should be subtracted from indicated inductance.

Test frequency range 100 through 200 MHz: For electrical characteristics measurements, use the HP190-A Q Meter, HP260-A Q Meter, HP4342A, HP250B RX Meter, or equivalent with the TF-405 test fixture, or equivalent (see figure 2). Fixture inductance (approximately 0.009 μH) should be subtracted from indicated inductance.

2/ Maximum current allowed is not to exceed the specified temperature rise.

3/ Applicable to ferrite core only.

TABLE II. Electrical characteristics (final).

| Inspection group | Allowable variation from initial measurements | | | |
|---|---|---|---|---|
| | Inductance (Percent) | DC resistance | Self-resonant frequency (Percent) | Q (Percent) |
| Qualification inspection | | | | |
| Group II | ±5 | ±(3% +.001 ohm) | -8 | -10 |
| Group IV | ±5 | ±(2% +.001 ohm) | -10 | -10 |
| Group V | ±2 | --- | --- | -10 |
| Conformance inspection group C | | | | |
| Subgroup II | ±5 | ±(3% +.001 ohm) | -8 | -10 |
| Subgroup IV | ±5 | ±(3% +.001 ohm) | -8 | -10 |



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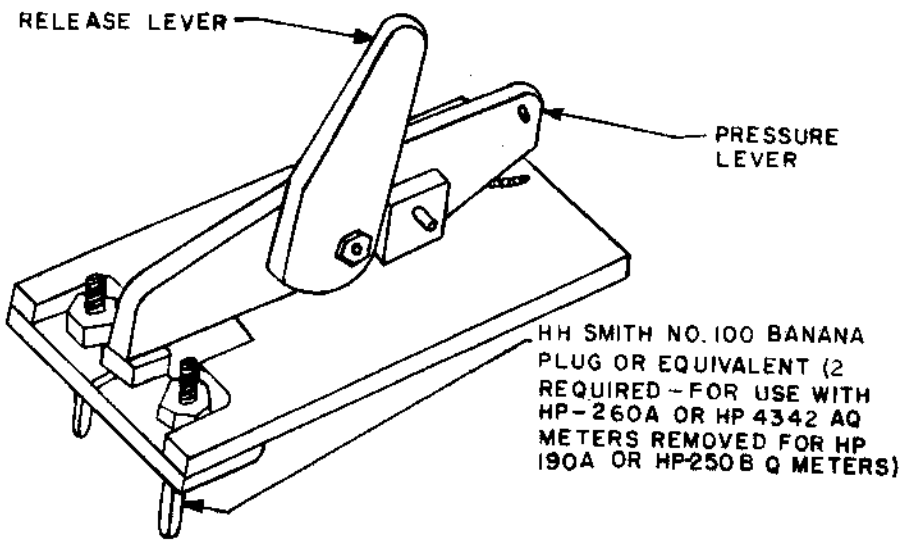



FIGURE 2. Typical chip coil test fixture (TF-405, or equivalent).

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-PRF-83446, this document references the following:

MIL-STD-202-105
MIL-STD-202-301

Custodians:
Army – CR
Navy - EC
Air Force – 85
DLA - CC

Review activities:
Army – MI
Navy – AS, OS
Air Force - 19

Preparing activity:
DLA – CC

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